AI-POWERED SMART INSURANCE CLAIM MANAGEMENT SYSTEM

With Fraud Detection and Automation

MCA Project

ARCHA DIJIKUMAR | MLM25MCA-2021

Department of Computer Applications

Mangalam College of Engineering, Ettumanoor

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Guide: Divya S B, HOD & Assistant Professor

Introduction

Background

Insurance claims often rely on paper forms, manual checks and slow approvals, delaying settlement for customers.

Purpose

Build a web-based system that digitises the claim journey and automatically flags suspicious claims.

Approach

Online claim submission, OCR document extraction, machine-learning fraud risk scoring and an approval workflow.

Importance

Faster settlement, fewer human errors, early fraud detection and real-time claim tracking for users.

Core idea: bring claim evidence, AI analysis and human verification into one connected workflow.

Abstract

Overview	An AI-powered web application that automates insurance claim processing and detects fraudulent claims.
Problem	Manual, paper-based processing is slow and error-prone; fraud is difficult to identify without intelligent detection.
Solution	Online submission, OCR-based document extraction, ML fraud-risk scoring, approval workflow and live tracking.
Technologies	Python, Django, MySQL, Scikit-learn and EasyOCR.
Outcome	Faster, more accurate and transparent claim handling, with every claim carrying a fraud-risk level before a decision.

Problem Definition

Manual, paper-based insurance claim processing is slow and error-prone. It gives users limited real-time tracking and cannot reliably detect fraudulent claims.

Manual flow

Long processing cycles and repeated handling

Human error

Mistakes during data entry and document checks

Fraud risk

Suspicious patterns may be missed

No tracking

Users lack timely visibility into claim progress

Objectives

Main objective

Develop a secure AI-powered web application to automate insurance claim processing and detect fraudulent claims.

- **Enable online claim submission**
- **Integrate OCR for document extraction**
- **Detect fraud using Machine Learning**
- **Classify claims by fraud risk**
- **Automate the claim approval workflow**
- **Provide real-time tracking and reports**

Existing System

Typical existing process



Slow processing

Paperwork and manual checks delay settlement.

Limited transparency

Claim status is not available in real time.

Operational cost

Repeated manual work increases effort and error risk.

Proposed System

Overview

A centralized AI-powered system for digital insurance claim submission, processing and monitoring.

How it solves the problem

Automates document processing and fraud detection, reducing manual work, processing time and errors.

Key features

Online claim submission; document upload with OCR; AI fraud detection; fraud-risk scoring; damage analysis; claim approval/rejection; real-time tracking.

Improvements

Faster processing, automated verification, better fraud identification, centralized data management and improved transparency for users and insurers.

Scope of the project

Target users

Policyholders
Insurance company staff
System administrators

Application areas

Motor insurance claims
Fraud detection
Document verification
Digital claim management

Functional coverage

Registration/login, claim submission, document OCR, fraud detection, damage analysis, tracking and approval/rejection.

Included in the project

Web-based user, insurance company and admin modules; centralized database; OCR document processing; fraud-risk scoring; claim status management.

Possible future applications

Real insurance-company integration, mobile apps, advanced image damage assessment, chatbot assistance, blockchain claim records and external insurance databases.

Literature Review

Title & Year	Author & Publication	Algorithm Used	Methodology	Result / Findings
Insurance Claim Prediction and Risk Assessment Using ML (2025)	Various Authors	Regression models	Claim prediction and customer risk assessment	Predicts claims, assesses risk; no automated fraud detection
Insurance Claims Estimation and Fraud Detection with Optimized Deep Learning (2025)	P. Anand Kumar, S. Sountharrajan	Deep learning	Claim estimation with optimized DL fraud detection	Detects fraud; high computational cost, complex
ML Based Method for Insurance Fraud Detection on Class Imbalance Datasets (2024)	A. A. Khalil, Z. Liu, A. Fathalla, A. Ali, A. Salah	ML with data preprocessing	Handles imbalanced claim data	Better fraud detection; lacks claim management
Fraud Detection and Analysis for Insurance Claim using ML (2022)	Abhijeet Urunkar	Random Forest, Decision Tree, SVM	Automatic fraud identification	Finds fraud automatically; no OCR or workflow
Insurance Fraud Detection: Evidence from AI and ML (2022)	F. Aslam, A. I. Hunjra, Z. Ftiti, W. Louhichi, T. Shams	Logistic Regression, SVM, Naïve Bayes	Feature selection and algorithm comparison	Identifies fraud features; no OCR or dashboard

Technology Stack

Frontend

HTML, CSS, JavaScript and Bootstrap through Django templates; Chart.js for admin dashboard charts.

Backend

Python with Django for server-side logic, views, URL routing, authentication and role-based access.

Database

MySQL stores user, company, claim, document, fraud and complaint data.

Machine Learning

Scikit-learn for fraud detection and fraud-probability prediction.

OCR

EasyOCR extracts text and key fields from uploaded documents.

Image processing

External AI image-analysis API (Gemini) analyses six uploaded vehicle damage images and generates a damage description.

Software and Hardware Requirements

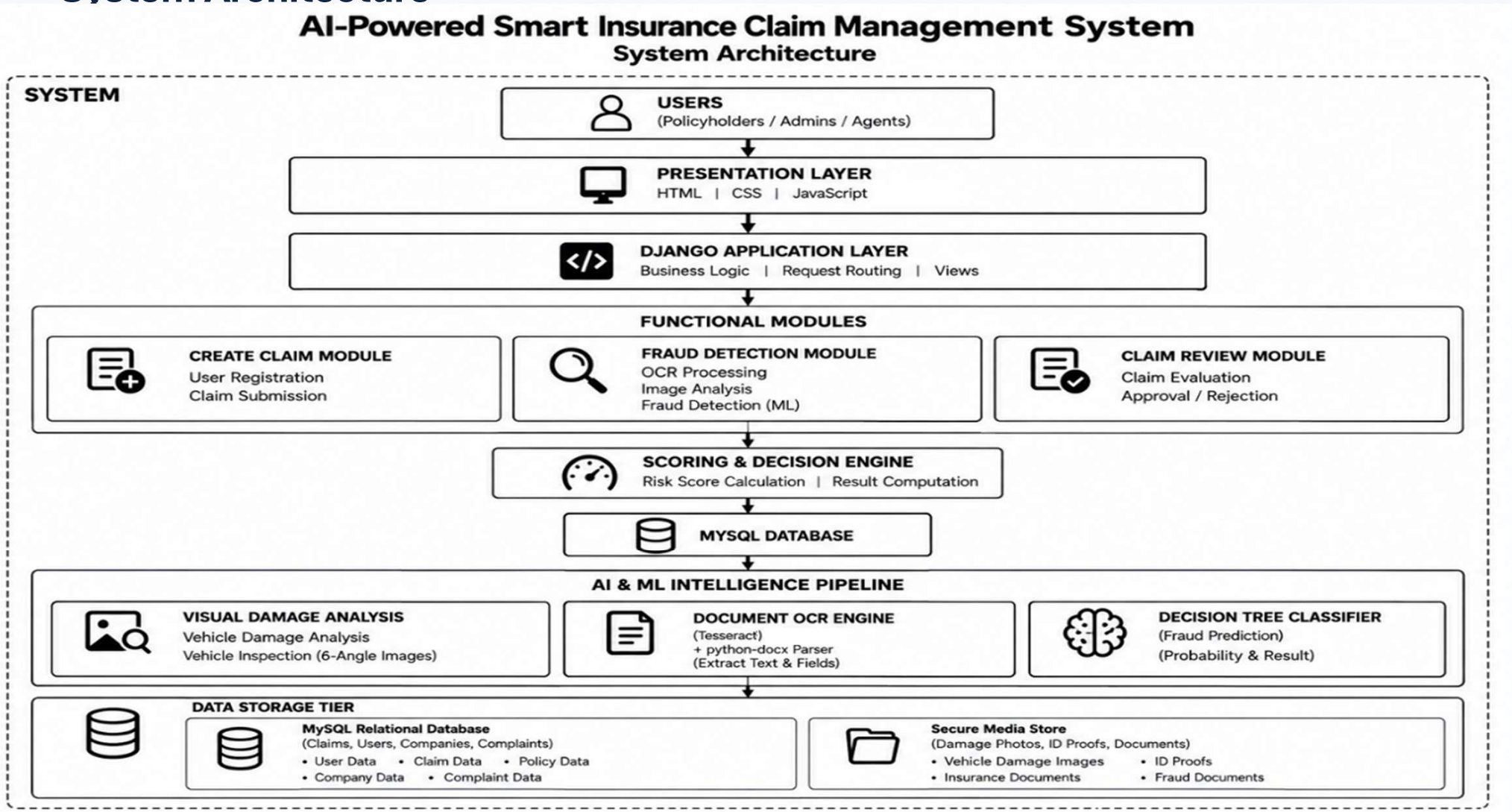
Software requirements

- Operating system: Windows 10/11 or Linux
- Frontend: HTML, CSS, JavaScript, Bootstrap (Django templates), Chart.js
- Backend: Python 3 with Django
- Database: MySQL
- ML / OCR: Scikit-learn, EasyOCR
- Image analysis: external AI image-analysis API (Gemini)
- Tools: VS Code, StarUML, Draw.io
- Browser: Chrome / Edge / Firefox

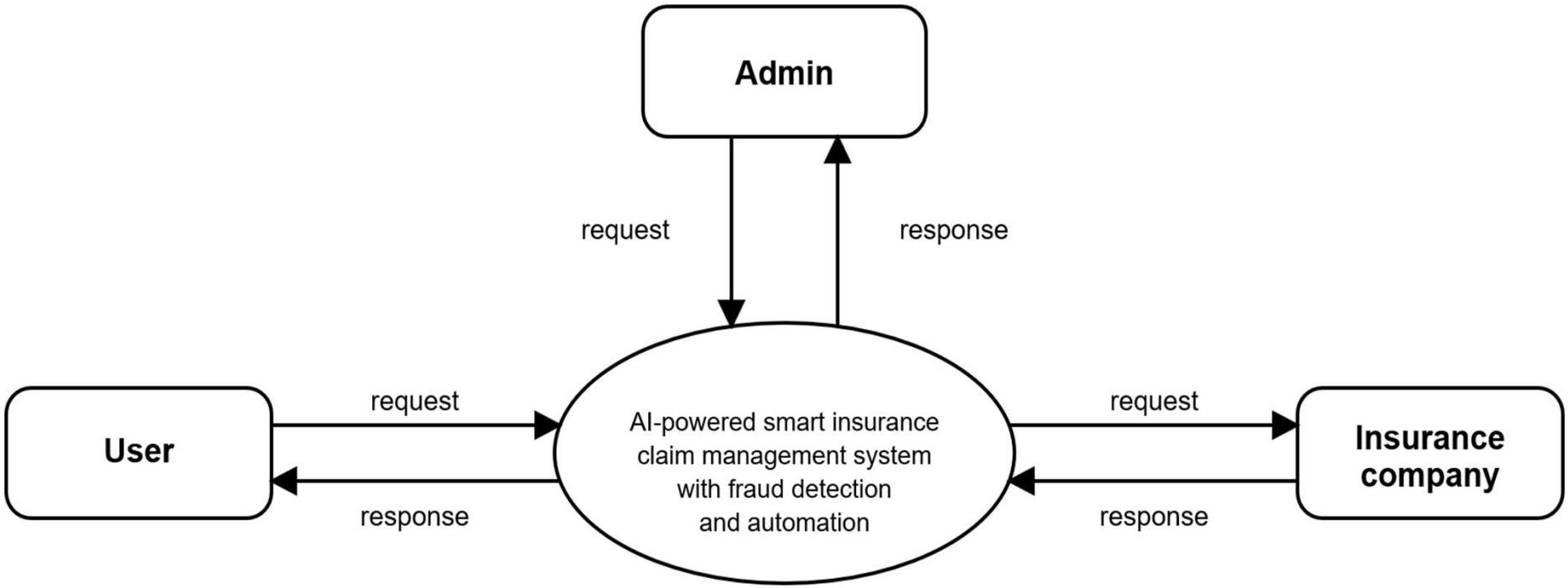
Hardware requirements

- Processor: Intel Core i5 / Ryzen 5 or higher
- RAM: 8 GB minimum
- Storage: 256 GB SSD or more
- Display: 1366 × 768 or higher
- Internet connection for the web application and external AI damage-analysis service

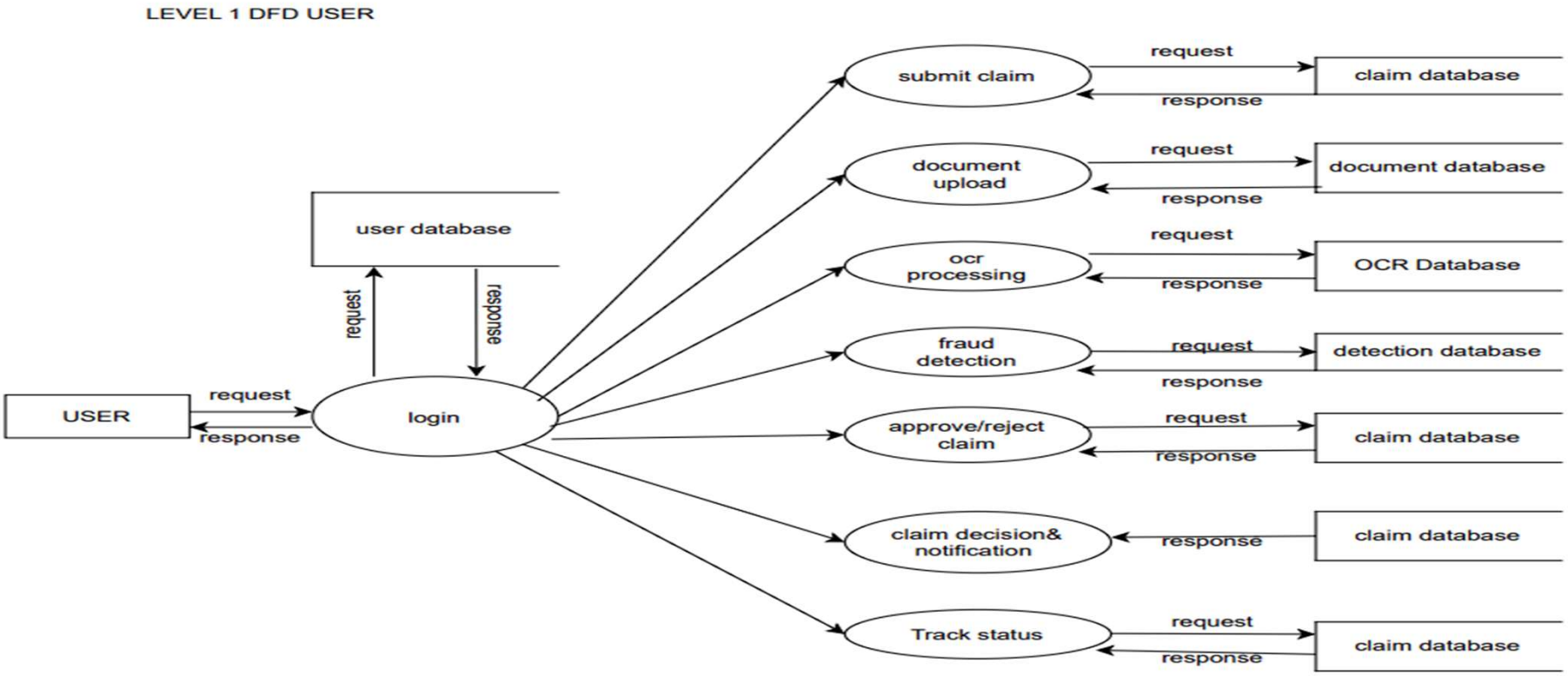
System Architecture



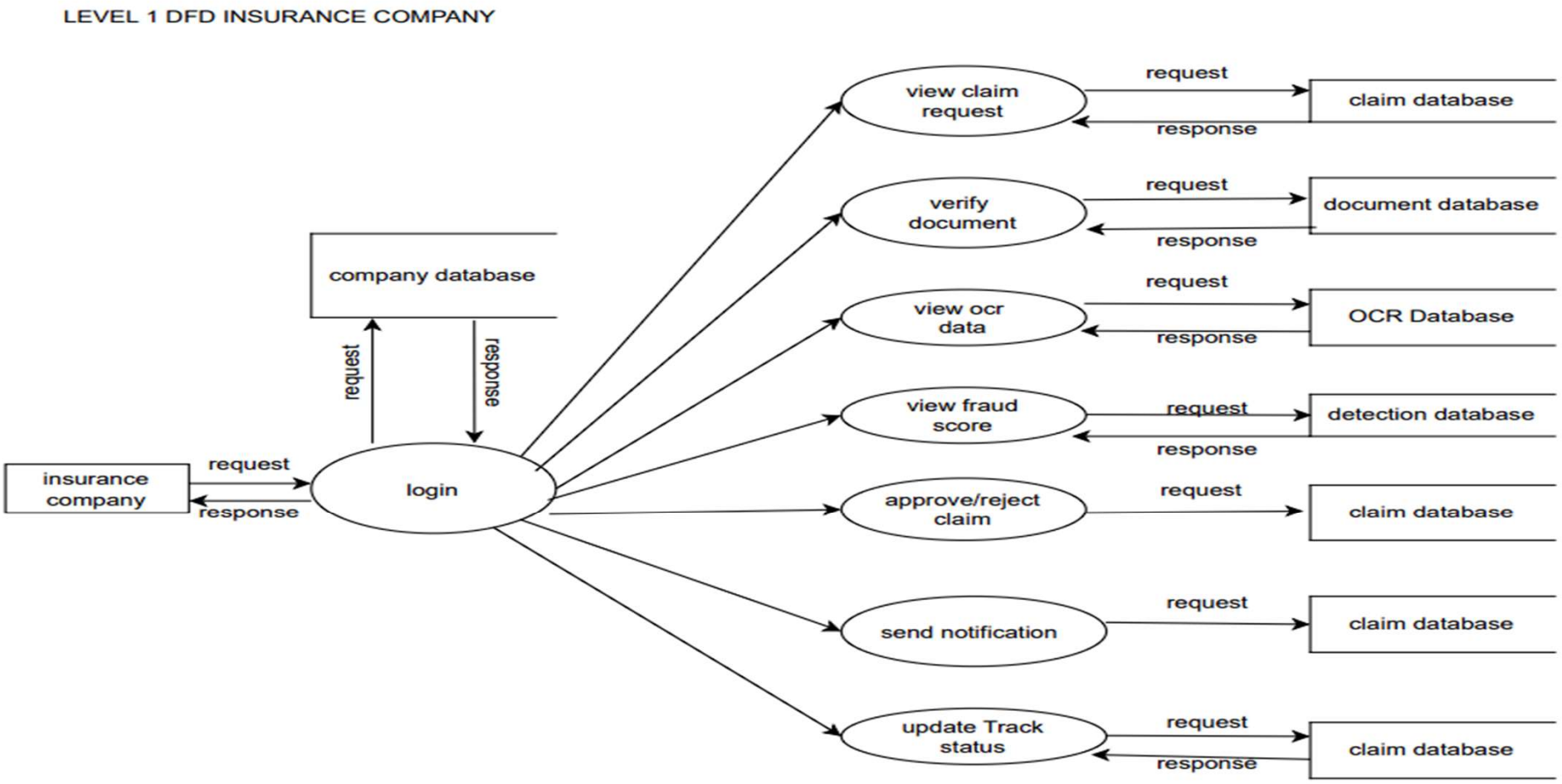
Data Flow Diagram — Level 0



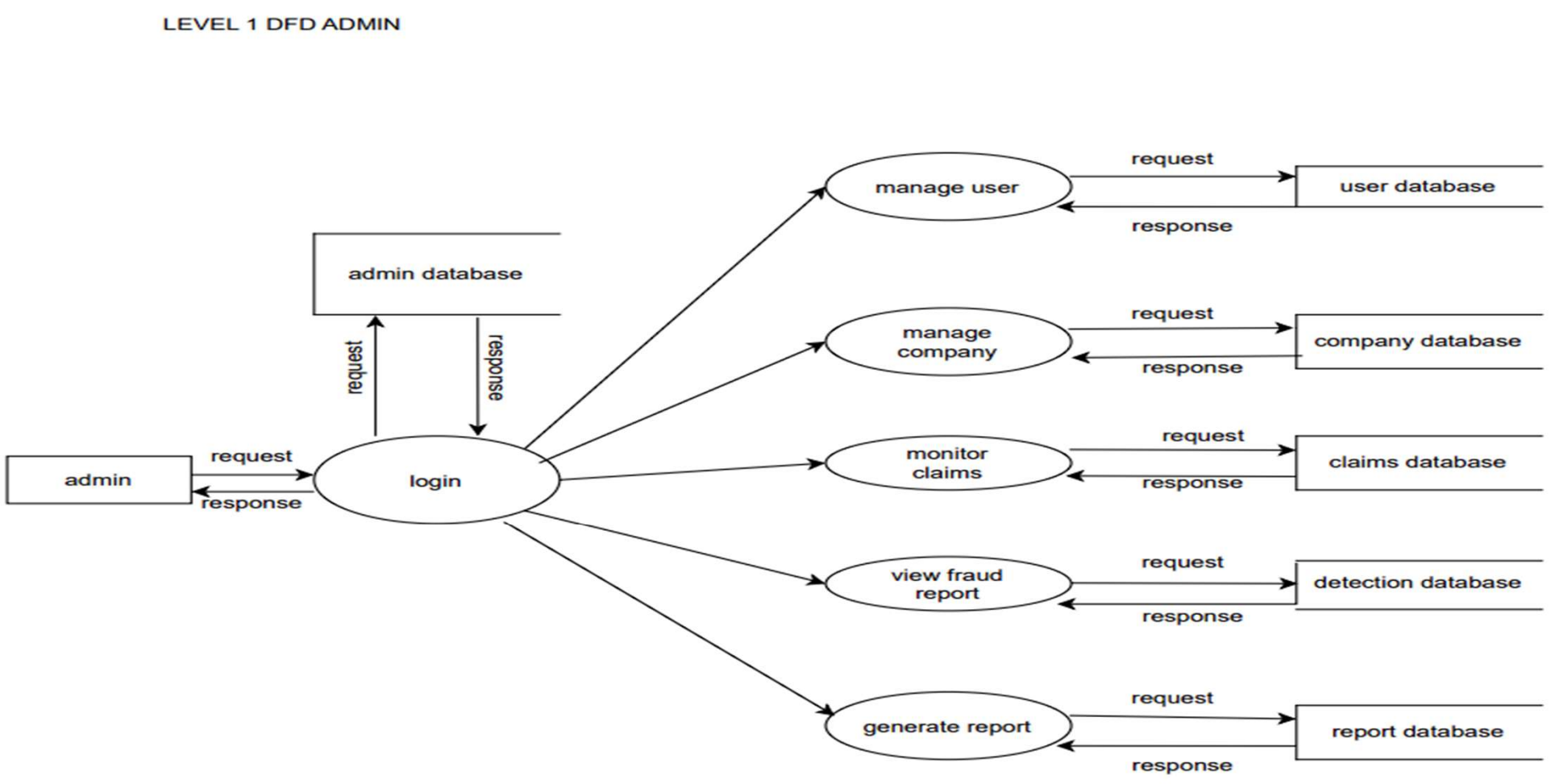
DFD Level 1 — User



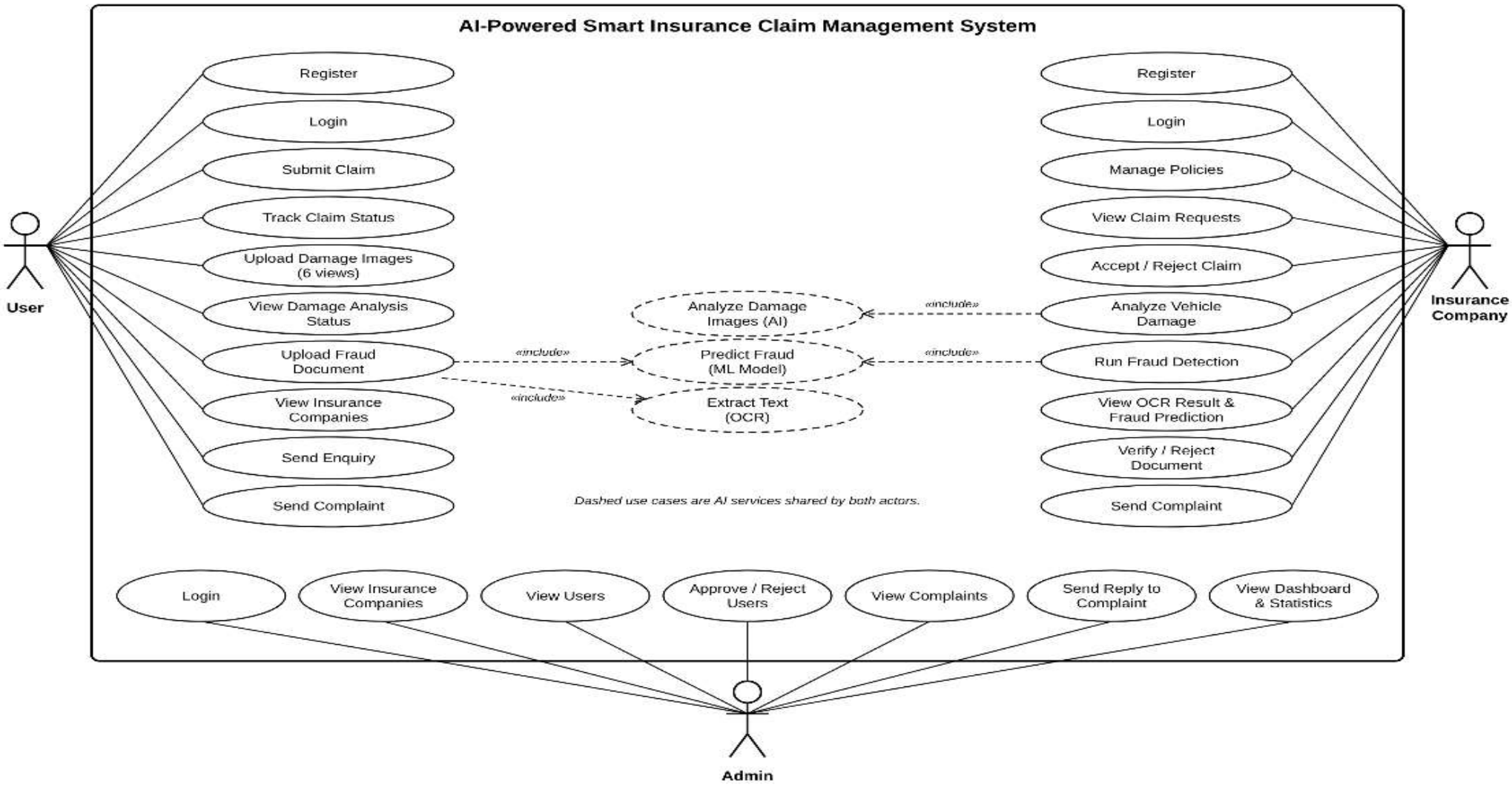
DFD Level 1 — Insurance Company



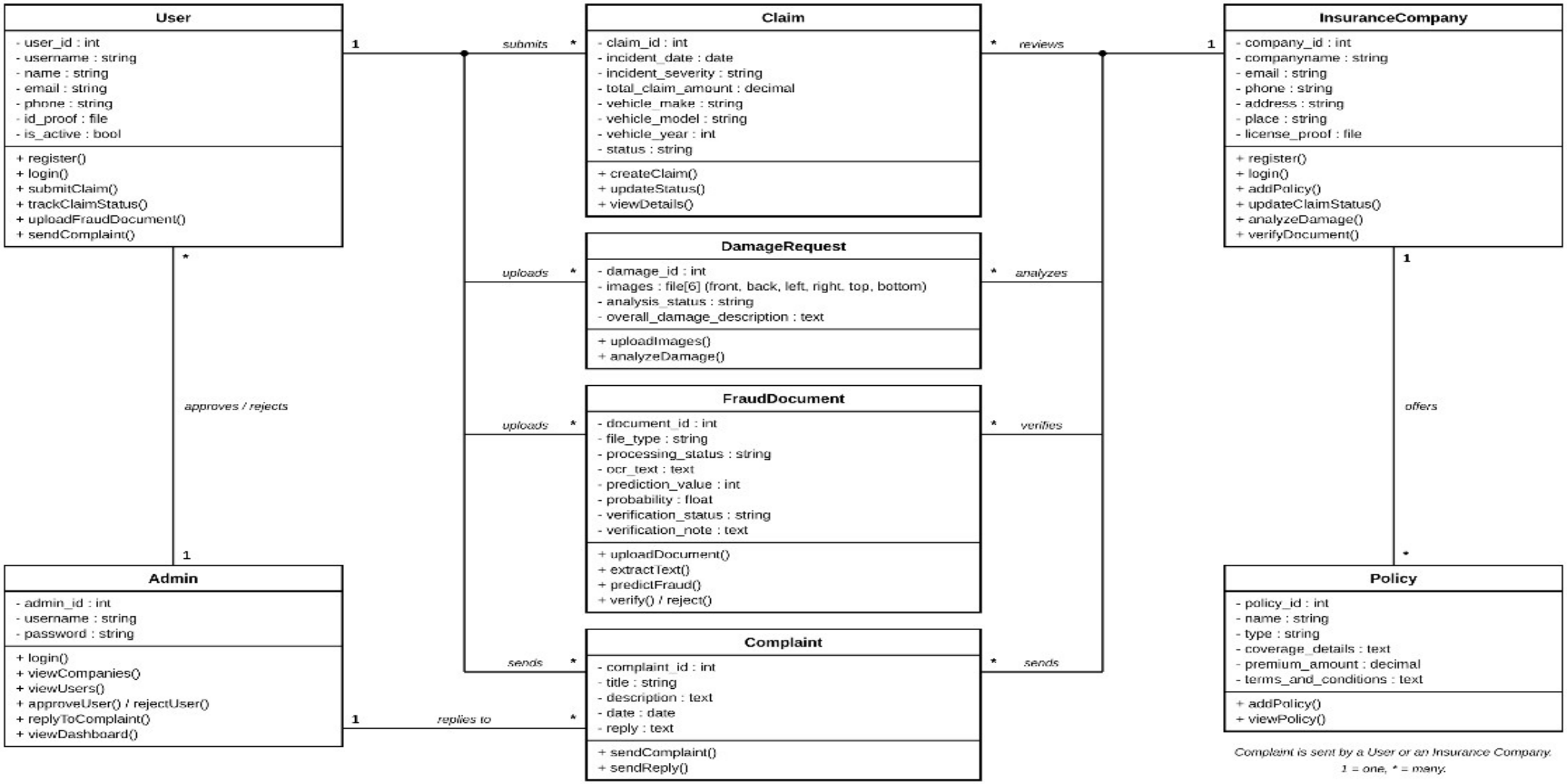
DFD Level 1 — Admin



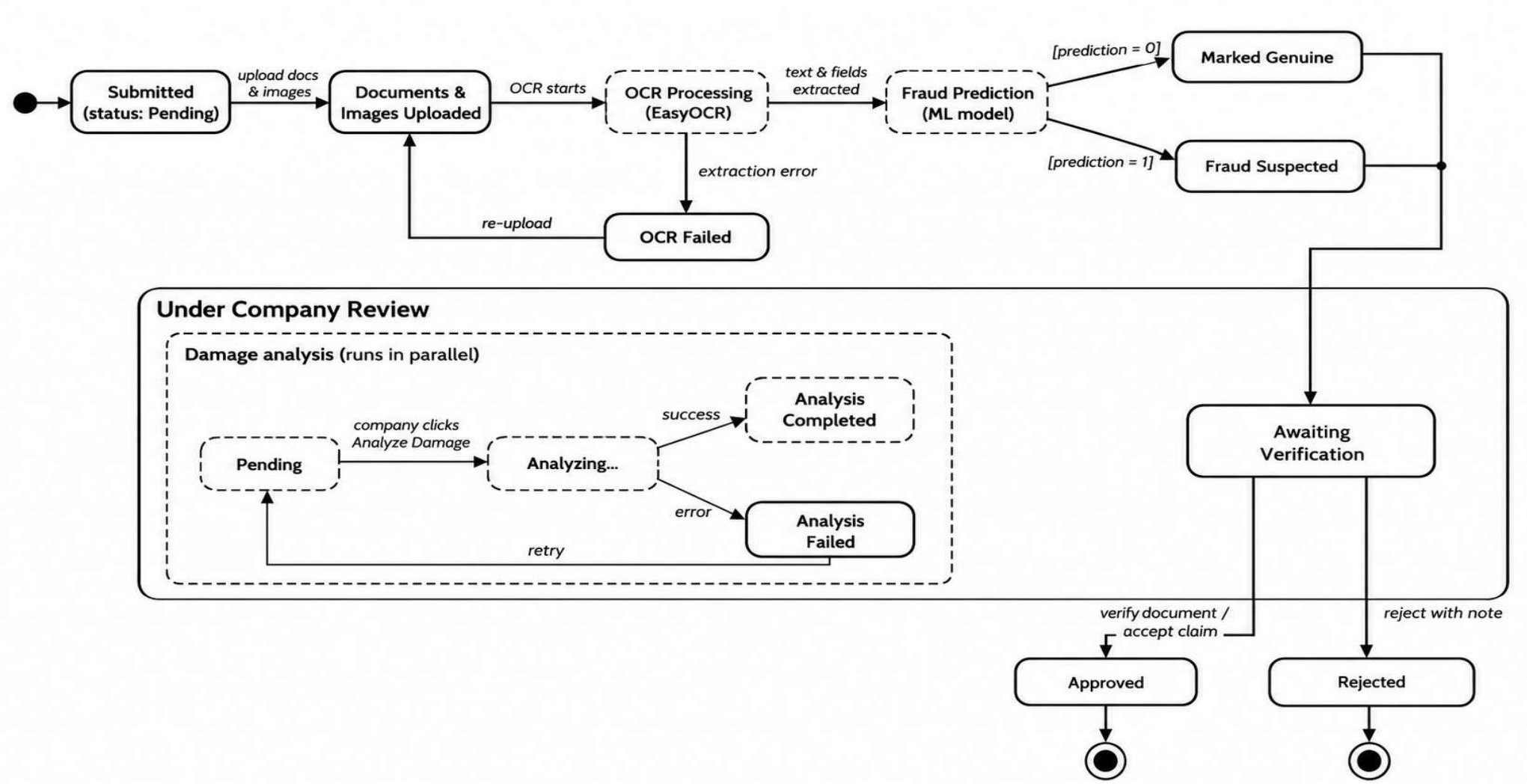
Use Case Diagram



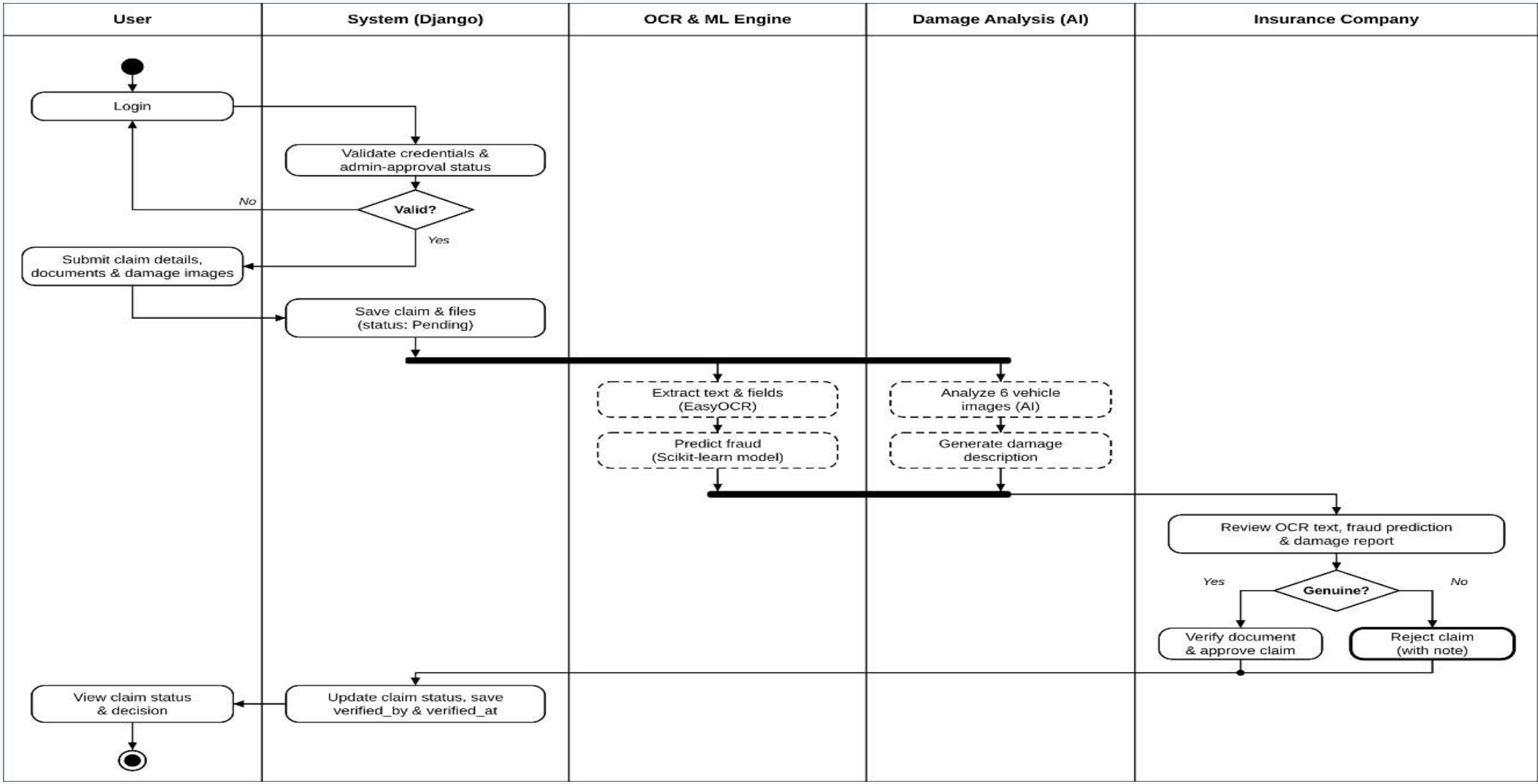
Class Diagram



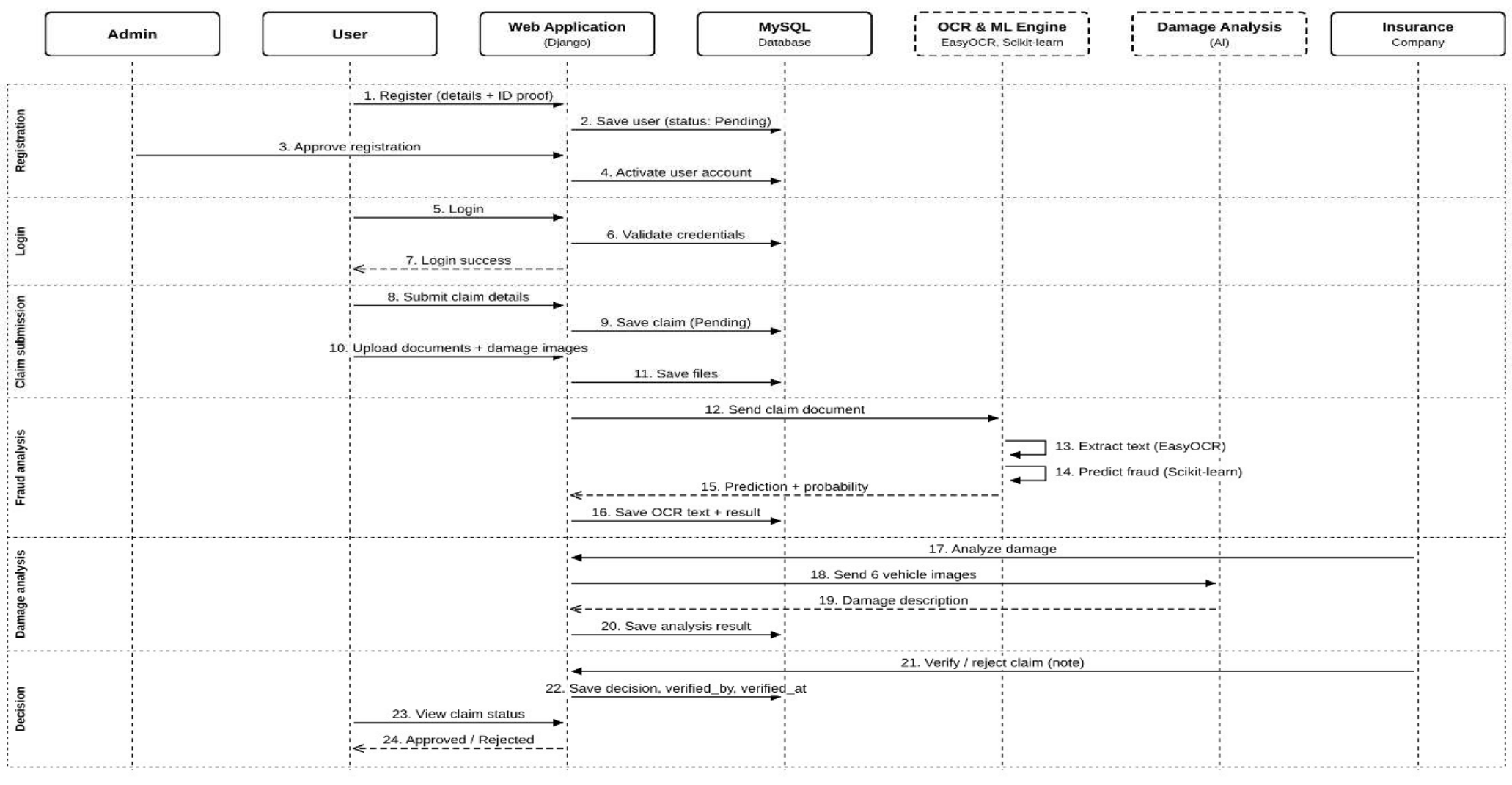
State Diagram



Activity Diagram



Sequence Diagram



Module Description

User Module

Register, submit claims, upload documents and track claim status.

OCR Module

Extract important information automatically from uploaded documents.

Fraud Detection Module

Use AI and machine learning to identify potentially fraudulent claims.

Image Processing Module

Analyze uploaded images to assess reported vehicle damage.

Insurance Company Module

Enable staff to review, verify, approve or reject claims.

Admin Module

Manage users, claims, complaints and other system activities.

Algorithm and Methodology

1

01 Submit

User registers, waits for admin approval, logs in and submits a claim with documents and six vehicle photos.

2

02 Extract

EasyOCR reads text and key fields from uploaded documents.

3

03 Analyse

AI image-analysis API describes damage in six photos; claim features are prepared for the model.

4

04 Score

Scikit-learn decision tree predicts fraud or non-fraud and gives a fraud probability as a risk indicator.

5

05 Decide

Insurance staff review results, verify or reject documents with a note, then approve or reject the claim. User tracks status live.

Development methodology: Agile, using iterative sprints, continuous testing and incremental feature delivery.

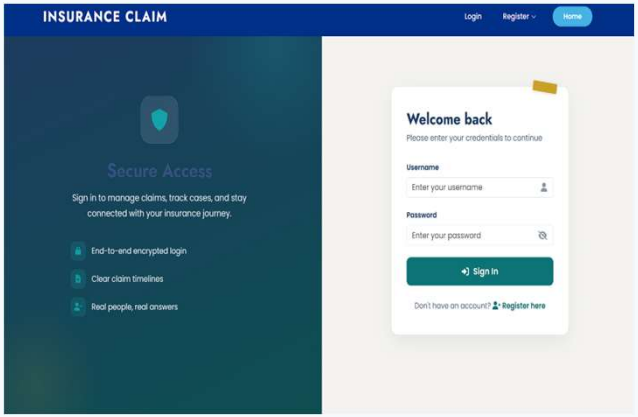
Implementation

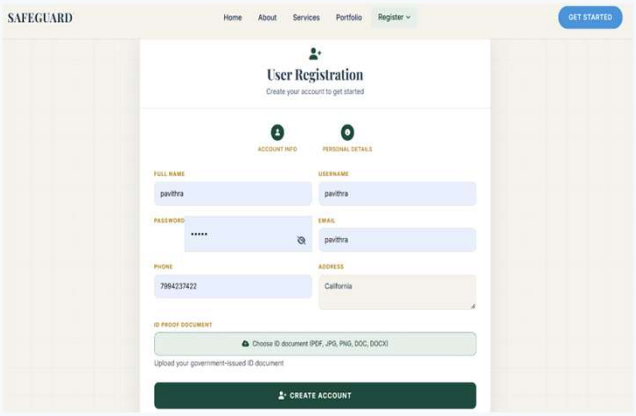
Frontend	HTML, CSS, JavaScript and Bootstrap interfaces for users, insurance staff and admin: registration, claim form, document/photo upload, status tracking and dashboards.
Backend	Python Django views and URL routing handle authentication, role-based access, policy and claim management, and admin approval workflow.
AI integration	EasyOCR extracts document data; Scikit-learn decision tree predicts fraud probability; Gemini image-analysis API describes damage in uploaded vehicle photos.
Database	MySQL stores users, companies, policies, claims, documents, fraud results and complaints in one centralized source.
Major functions	Authentication, claim submission, uploads, OCR extraction, fraud-risk scoring, damage analysis, approval/rejection and tracking.

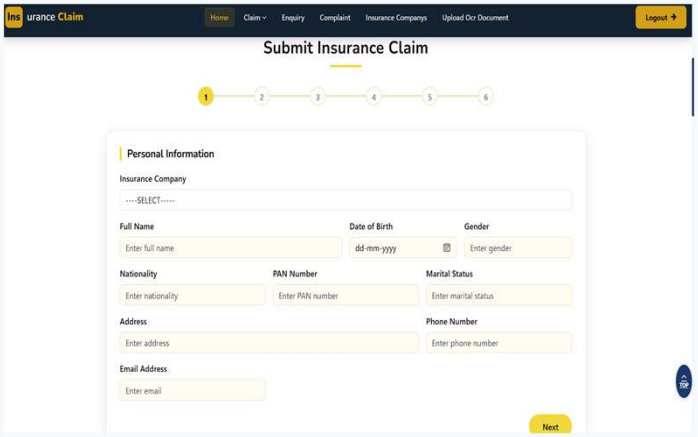
Result Analysis

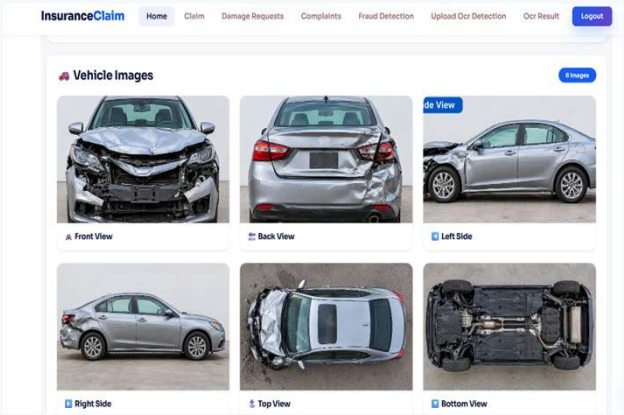
Test case	Result	Observation
Registration and approval	**Pass**	The system gives a fraud probability used as a risk indicator, not a final decision.
Claim submission and validation	**Pass**	OCR and damage analysis work best with clear documents and photos.
Fraud prediction	**Pass**	All modules work together from claim submission to final decision.
OCR extraction	**Pass**	Document text and key fields can be extracted.
Damage analysis	**Pass**	Damage description is generated from uploaded vehicle images.

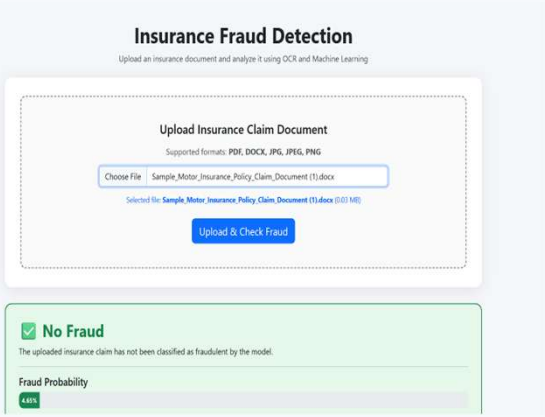
Screenshots / Output Evidence

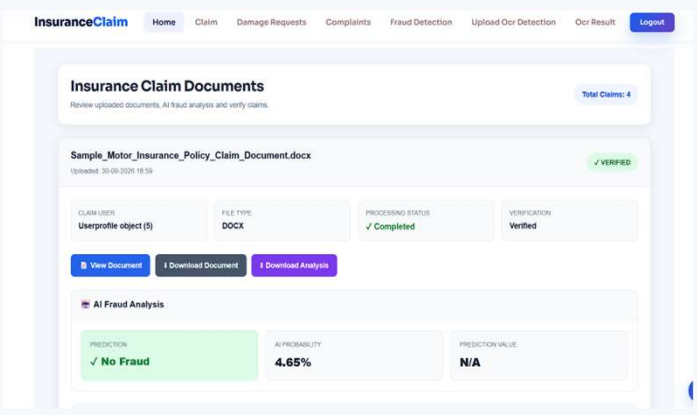












Comparison with Existing System

Feature	Existing system	Proposed system
Processing	Manual, paper-based	Automated online claim submission and review
Verification	Manual document checking	OCR-based text and field extraction (EasyOCR)
Fraud detection	Manual; depends on reviewer experience	ML prediction with a fraud probability
Damage assessment	Physical inspection or manual photo review	AI-generated damage description from six vehicle photos
Time	Slow due to paperwork	Faster review with evidence in one place
Decision record	Scattered across files and emails	Verification note, verifier and time stored with the document

Testing

Unit testing

Each function tested in isolation: registration, login, claim submission, document upload and fraud prediction.

Integration testing

Django views, MySQL, OCR and ML model exchange data correctly.

System testing

End-to-end claim flow tested from registration through final decision.

Validation and role testing

Invalid inputs are rejected; Admin, Insurance Company and User sessions are tested separately.

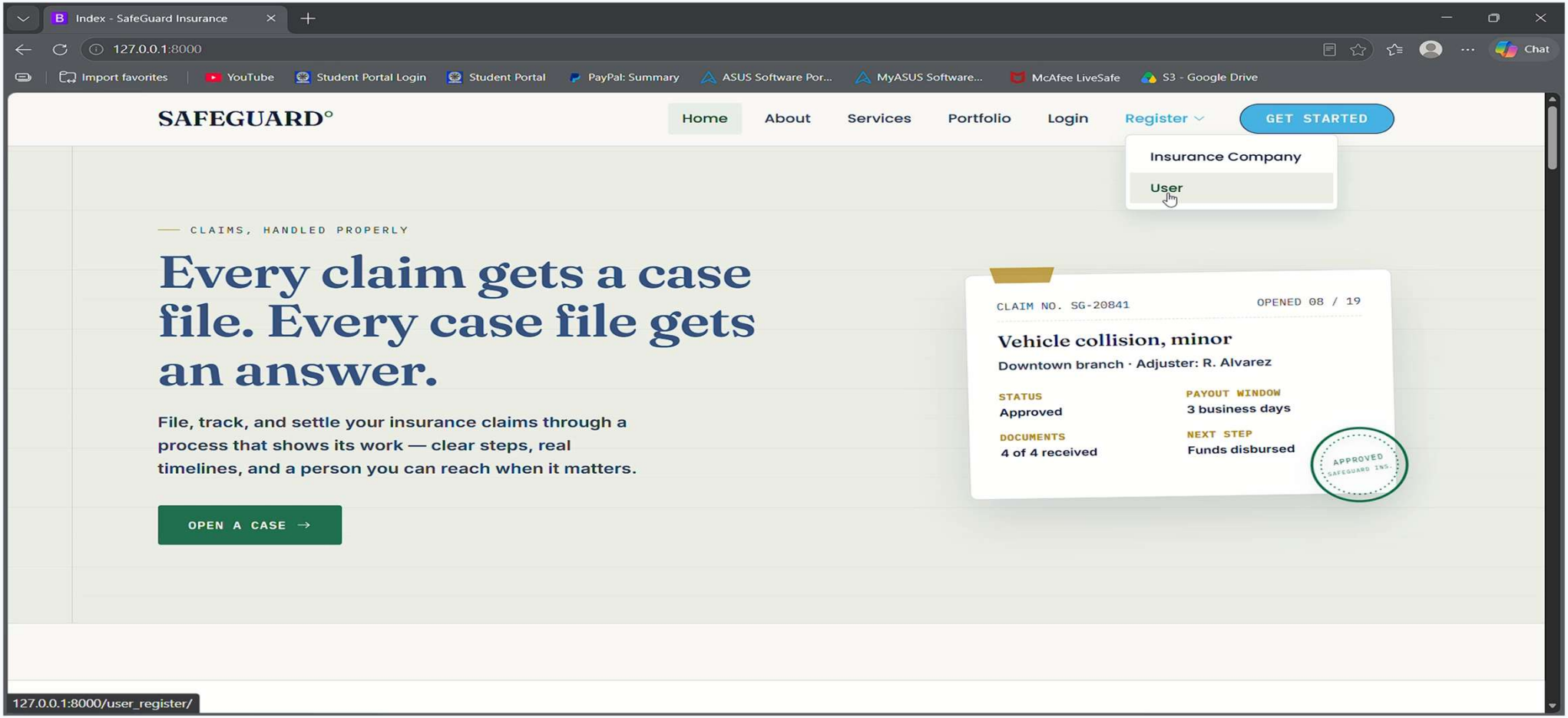
Recorded test outcomes

Valid login: pass; invalid login: denied; claim saved with six images and Pending status; OCR text/fields saved; fraud prediction generated; approve/reject status updated and visible.

Testing

Test case	Expected result	Actual result	Status
Valid login	Login successful	—Login successful	—Pass
Invalid login	Error message shown	—Login denied	—Pass
Claim submission	Claim saved with all fields and six images, status Pending	—Claim saved	—Pass
OCR upload	Text extracted and key fields saved	—Text and fields saved	—Pass
Fraud prediction	Fraud / non-fraud prediction with probability generated	—Prediction generated	—Pass
Approve / reject	Status updated and visible to the user	—Status updated	—Pass

Demo Video


Index - SafeGuard Insurance
127.0.0.1:8000
Chat
Import favorites
YouTube
Student Portal Login
Student Portal
PayPal: Summary
ASUS Software Por...
MyASUS Software...
McAfee LiveSafe
S3 - Google Drive
SAFEGUARD°
Home
About
Services
Portfolio
Login
Register
GET STARTED
Insurance Company
User
CLAIMS, HANDLED PROPERLY
Every claim gets a case file. Every case file gets an answer.
File, track, and settle your insurance claims through a process that shows its work — clear steps, real timelines, and a person you can reach when it matters.
OPEN A CASE →
CLAIM NO. SG-20841
OPENED 08 / 19
Vehicle collision, minor
Downtown branch · Adjuster: R. Alvarez
STATUS
Approved
PAYOUT WINDOW
3 business days
DOCUMENTS
4 of 4 received
NEXT STEP
Funds disbursed
APPROVED
SAFEGUARD INS.
127.0.0.1:8000/user_register/

Advantages

Faster processing

Online submission removes paperwork; reviewers see evidence in one place.

Fewer errors

OCR reduces manual data-entry and document-checking mistakes.

Fraud detection

ML provides a fraud prediction with probability, helping flag suspicious claims early.

Transparency

Users track claim status on their dashboard without visiting the office.

Centralized data

One database stores users, companies, claims, documents and analysis results.

Low cost

Uses open-source technologies: Python, Django, MySQL, Scikit-learn and EasyOCR.

Limitations

OCR quality

Extraction accuracy depends on the clarity of uploaded documents.

Model quality

Fraud predictions depend on training data: 1,000 claim records, with fewer fraud cases (247) than genuine cases (753).

Human review

The model gives a prediction only; the insurance company still makes the final verification and rejection/approval decision.

Connectivity

An internet connection is required for the web application and external AI damage-analysis service.

Conclusion

The AI-Powered Smart Insurance Claim Management System replaces slow manual processing with an automated, transparent workflow that combines OCR, AI-based damage analysis and machine learning to detect fraud—while leaving the final decision with the insurance company.

Automated

Claim submission to approval is handled online, with OCR reading documents.

Intelligent

Each claim gets a fraud prediction with probability, and vehicle photos receive an AI damage description.

Transparent

Users see real-time claim status; admin and company dashboards support oversight and reports.

AI supports decision-making; it does not replace human verification.

Future Scope

Mobile app

Android and iOS applications for claim filing on the go.

Deep learning

Stronger fraud and damage detection models.

Advanced damage analysis

Estimate repair cost, compare it with claimed amount, and detect reused or manipulated images.

Notifications

Email/SMS alerts and a support chatbot.

More insurance types

Extend to health, life and property insurance with separate claim forms and fraud models.

References

- [1] P. Anand Kumar and S. Sountharrajan, “Insurance claims estimation and fraud detection with optimized deep learning techniques,” Scientific Reports, 2025, doi: 10.1038/s41598-025-12848-0.
- [2] A. A. Khalil, Z. Liu, A. Fathalla, A. Ali, and A. Salah, “Machine learning based method for insurance fraud detection on class imbalance datasets with missing values,” IEEE Access, vol. 12, 2024, doi: 10.1109/ACCESS.2024.3468993.
- [3] A. Urunkar et al., “Fraud detection and analysis for insurance claim using machine learning,” Proc. 2022 IEEE Int. Conf. Signal Processing, Informatics, Communication and Energy Systems (SPICES), Mar. 2022, pp. 406–411.
- [4] F. Aslam, A. I. Hunjra, Z. Ftiti, W. Louhichi, and T. Shams, “Insurance fraud detection: Evidence from artificial intelligence and machine learning,” Research in International Business and Finance, vol. 62, Art. no. 101325, 2022.
- [5] M. Shanthini, “Stacking classifier-based automated insurance fraud detection system,” Proc. 2022 IEEE Conf. Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI), 2022, pp. 1–6.

References are transcribed from the source presentation.

THANK YOU

AI-Powered Smart Insurance Claim Management System

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